# Severity-Aware Conformal Prediction

How to Weight Probability and Loss

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## Goal

Class-conditional conformal prediction bounds how often adverse cases pass human review unchecked, but not how costly they are. We seek the review rule with the fewest reviews that controls both.

$$\min_{\text{rule}} \Pr(\text{review}) \quad \text{subject to} \quad \underbrace{\Pr(\text{pass} \mid Y=1) \le \alpha_{\text{cov}}}_{\text{frequency}}, \qquad \underbrace{\mathbb{E}\big[C\,\mathbf{1}\{Y=1,\ \text{pass}\}\big] \le \alpha_{\text{sev}}}_{\text{severity}}$$

## Contributions

### 1. The optimal rule (Theorem 1)

With default probability $p$ and expected loss given default $c(x) = \mathbb{E}[C \mid Y=1, X=x] \le B$, the optimal rule passes $x$ when $h_{\theta^*}(x) < t^*$, where

$$h_\theta(x) = (1-\theta)\,p(x) + \theta\,\frac{p(x)\,c(x)}{B}, \qquad \theta^* = \begin{cases} 1 & \text{only severity binds} \\ \text{strictly decreasing in } \alpha_{\text{sev}} & \text{both bind} \\ 0 & \text{only coverage binds.} \end{cases}$$

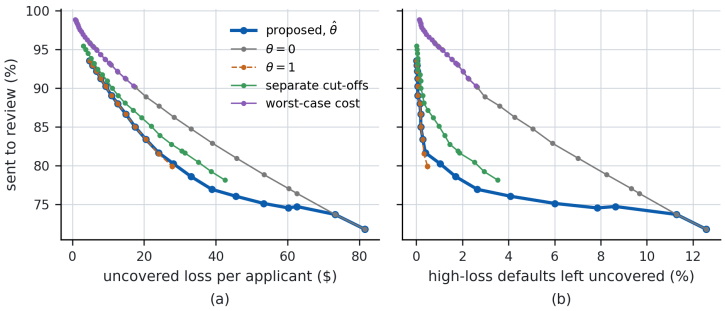
### 2. Estimation with exact guarantees (Theorem 2)

The weight comes from a plug-in dual linear program on $n_U$ unlabeled cases, without outcomes.

$$\max_{\phi \in [0,1]^{n_U}} \sum_i \phi_i \ \text{ s.t. } \ \sum_i \phi_i \hat{p}_i \le \alpha_{\text{cov}} \sum_i \hat{p}_i, \ \ \sum_i \phi_i \hat{p}_i \hat{c}_i \le n_U \alpha_{\text{sev}}, \qquad \hat{\theta} = \frac{B\hat{\lambda}_{\text{sev}}}{\hat{\lambda}_{\text{cov}} + B\hat{\lambda}_{\text{sev}}}$$

Multi-risk conformal risk control calibrates $\hat{t} = \min(\hat{t}_{\text{cov}}, \hat{t}_{\text{sev}})$ on $h_{\hat{\theta}}$, so both constraints hold in finite samples for any $\hat{\theta}$ that does not depend on the calibration sample. The weight error is $O_P(n_U^{-1/2})$, and the excess review vanishes as the samples grow and $\hat{p}$, $\hat{c}$ improve.

## Evidence




| At $\alpha_{\text{sev}} = \$60$ | Review (%) | Cov. used | Sev. used |
|---|---|---|---|
| **Proposed** | **74.5** | 85 | 100 |
| $\theta = 0$ | 77.0 | 73 | 100 |
| $\theta = 1$ | 79.9 | 101 | 46 |
| Separate cut-offs | 81.9 | 59 | 51 |
| Worst-case cost | 93.3 | 12 | 17 |

LendingClub loan dataset, 283,026 loans issued in 2015, $\alpha_{\text{cov}} = 0.10$, means over 100 random splits. The last two columns give the share of each level used (%).

- **Fewer reviews.** No more reviews than the four other rules at 20 of 21 severity levels (left panel). The other rules leave much of one level unused (table) and send more applicants to review than the levels require.
- **Theory holds in practice.** $\hat{\theta}$ follows the three regimes of Theorem 1 on real loans. In simulations its error shrinks like $n_U^{-1/2}$, and the excess review falls from 11.7 to 0.5 percentage points as the samples grow.
- **Calibration is needed.** Setting the cut-off from $\hat{p}$ and $\hat{c}$ alone saves 1.9 percentage points of review at \$60 but exceeds the severity level at 19 of 21 levels, by up to 20%.

## Limitations and next steps

- **Biased estimates.** $\hat{p}$ and $\hat{c}$ understate 2015 losses, so $\hat{\theta}$ falls below the best weight, costing up to 2.1 percentage points of review.
- **Calibration delay.** Calibrating on loans issued before 2015 exceeds both levels.
- **Richer constraints.** Tail losses instead of expected losses, and more than two constraints, where the weights need not be monotone.